

IEM KOLKATA**Calculus
Integration**

Q1) $\int_1^3 x\sqrt{x^2-1}dx =$

A. $\frac{8\sqrt{8}}{3}$ B. $\frac{\sqrt{8}}{3}$ C. $\frac{8}{3}$ D. None of the above

Q2) $\int_0^2 |1-x| dx$

A. 0 B. 1 C. 2 D. None of the above

Q3) $\int_{-\pi}^{\pi} (|x| + \sin x) dx =$

A. π B. π^2 C. 2π D. $-\pi$

Q4) $\int_0^{\frac{\pi}{2}} \log(\tan x) dx$

A. 0 B. 1 C. $\sec^2 x$ D.

Q5) $\int_{-\pi}^{\pi} \log\left(\frac{1+\sin x}{1-\sin x}\right) dx =$

A. 0 B. 1 C. π D. $-\pi$

Q6) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x \cos x) dx =$

A. 0 B. 1 C. 2 D. $\frac{4}{3}$

Q7) $\int_{-1}^1 \sqrt{\frac{1+x}{1-x}} dx =$

A. π B. π^2 C. 2π D. $-\pi$

Q8) The value of the integral $\int_0^{\infty} e^{-y^3} y^{\frac{1}{2}} dy$

A. $\sqrt{\pi}$ B. $\frac{\sqrt{\pi}}{2}$ C. $\frac{\sqrt{\pi}}{3}$ D. $\frac{\sqrt{\pi}}{4}$

Q9) Evaluate $I = \int_0^1 \frac{x}{(1+x^2)} dx$

A. 0 B. 1 C. $\ln 2$ D. $\frac{1}{2} \ln 2$

Q10) The value of $\int_0^\pi x \sin x dx$ is

A. 0 B. 1 C. π D. 2π

Q11) $\int_0^1 x \ln x dx$

A. $-\frac{1}{4}$ B. 1 C. 2 D. Cannot be determined

Q12) The value of $\int_0^\pi x^2 \cos x dx$ is

A. -2π B. $-\pi$ C. π D. 2π

Q13) Find the value of $\int_0^{\frac{\pi}{2}} \frac{\tan x}{\tan x + \cot x} dx$

A. $\frac{\pi}{2}$ B. $\frac{\pi}{3}$ C. $\frac{\pi}{4}$ D. π